IDA117V Reproducibility Appendix

Predicting Seven-Day Municipal Service-Request Non-Completion Risk

This appendix records the exact source snapshot, execution protocol and model evidence produced by the repository package. It supplements, rather than replaces, the compiled assignment report.

Target repository: https://github.com/mahlatseselowa/IDA117V.git. The final target-repository commit hash is not stated here because it must be read from that exact repository after the package is committed.

Dataset snapshot

Item	Code-derived value
Source rows / columns	247,305 / 14
Rows excluded: missing creation time	131
Model rows	247,174
Date coverage	2026-04-01 to 2026-05-30
Within seven days	152,437
At risk	94,737 (38.3%)
Train / validation / test	168,427 / 43,234 / 35,513
Source SHA-256	ca6e1f6a2648e33bf0ee0cf2f269544be705c26051c79c82a6544b7487dfd0fc

Locked-test result

Metric	Optimized LightGBM
Decision threshold	0.65
Accuracy	0.6459
Macro-F1	0.6458
At-risk precision	0.5815
At-risk recall	0.7073
ROC-AUC	0.7273
Average precision	0.7227

Execution order

1. Install the exact versions in requirements.txt using Python 3.12.
2. Place the source workbook in 02_Data/raw; it is excluded from Git pending licence confirmation.
3. Run 03_Code/run_analysis.py to regenerate the model, figures and result evidence.
4. Run 03_Code/make_documentation.py to refresh the data dictionary and snapshot note.
5. Run 03_Code/build_report.py to regenerate this appendix.
6. Run 03_Code/verify_interface.py and then launch 03_Code/app.py with Streamlit.

Interpretation limits

- The seven-day horizon and creation-time availability of Work_Center require stakeholder confirmation.
- The extract covers only two months; seasonal and longer-term generalisation is not established.
- Missing completion is treated conservatively as at risk; source-system status semantics must be confirmed.
- Location fields may be quasi-identifiers. The model must not be used to deny, close or automatically deprioritise a request.
- Predictions are associations, not causal findings. Human review remains mandatory.

Snapshot consistency

Counts are computed from the workbook packaged here and may differ from earlier report drafts that used a different CSV snapshot.

Saved evidence inventory

- dataset_audit.json and run_summary.json - source fingerprint, split and headline metrics.
- validation_model_comparison.csv and test_model_comparison.csv - identical-case model comparison.
- lightgbm_trials.csv, ablation_results.csv and learning_curve.csv - controlled selection and diagnosis.
- heldout_predictions.npz and locked_test_predictions.csv - paired locked-test evidence.
- subgroup_audit.csv and capacity_scenarios.csv - fairness/operations review inputs.
- interface_ui_tests.json - three valid cases, one invalid case and HTTP health check.

Submission and approval status

The table distinguishes completed package evidence from items that still require external evidence. No approval, public link, signature or repository commit is claimed without support.

Item	Status	Evidence still required / control
Manual-upload repository package	Completed and verified locally	Target URL supplied as https://github.com/mahlatseselowa/IDA117V.git; public accessibility and final commit hash remain unverified until upload/commit.
Dataset licence and redistribution	Recorded metadata only; approval not claimed	Exact item-level Custom License text, source-to-file chain of custody and written redistribution permission. Raw data remains excluded.
Entity-embedding architecture	Implemented; lecturer approval required and not evidenced	Dated lecturer email, LMS message, signed note or other explicit approval, or replace with an accepted architecture.
Streamlit deployment	Local launch and health check verified	Public app link is not created or verified.
Presentation/recording	Plan and local demo evidence completed	Public or lecturer-accessible recording link is not created or verified.
Contribution statement	One-page blank template included	Real names, student numbers, roles, specific outputs, dates and signatures.

McNemar interpretation

McNemar's test in this project evaluates paired correctness on the same locked cases. It tests whether the two discordant probabilities - champion only correct versus comparator only correct - are equal. It does not test macro-F1 or the difference between macro-F1 scores. Macro-F1, precision and recall remain separate descriptive metrics for practical interpretation.

Streamlit demonstration evidence

Case	Date	Complaint	Probability	Expected / verified outcome
Example 1	2026-05-23	Suspect Tampering	79.7%	At risk; observed at risk; passed
Example 2	2026-05-28	Meter: Damaged / Faulty	68.2%	At risk; observed at risk; passed
Example 3	2026-05-30	Customer Reconnection Request	51.6%	Below 0.65; observed within 7 days; passed
Invalid	-	Blank complaint type	-	Complaint type is required; passed

Local launch and HTTP 200 are verified. Public Streamlit link: [NOT CREATED / NOT VERIFIED]. Presentation/recording link: [NOT CREATED / NOT VERIFIED].

Target-repository hand-off

Commit the repository-ready package only after access to the exact target repository has been established. The raw workbook is intentionally excluded by .gitignore pending licence confirmation. After commit, verify the remote URL, a clean working tree and the 40-character HEAD hash. Never substitute the reference repository or reuse its commit hash.

Responsible-use conclusion

The evidence supports only a cautious human-controlled pilot - not automated denial, closure, deprioritisation, staff discipline or public ranking. Revalidate after material change.